Lie Theory:Assignment 5

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Problem 1

Find Weyl groups of $\mathrm{A}_2, \mathrm{B}_2, \mathrm{G}_2$.

Solution: Suppose the root system we want to discuss has simple roots β_1, β_2 with angle θ, and let its Weyl group be denoted by W. Let $\theta = \pi - \pi/m$. Let s_1, s_2 denote the reflections corresponding to β_1, β_2 respectively. According to our textbook, the Weyl group can be generated by the reflections of the simple roots; therefore, we just need to clarify the relations between s_1 and s_2.

For A_2, since $\theta = 2\pi/3$ and $m = 3$, we then have:

$$s_1(\beta_1) = -\beta_1 \quad s_1(\beta_2) = \beta_1 + \beta_2 \quad s_2(\beta_1) = \beta_1 + \beta_2 \quad s_2(\beta_2) = -\beta_2.$$

Then we obtain that $(s_1 s_2)^3 = \mathrm{id}$. Therefore, the Weyl group of A_2 is

$$W = \langle s_1, s_2 \mid s_1^2 = s_2^2 = (s_1 s_2)^3 = \mathrm{id} \rangle$$

which is exactly the Dihedral group D_3.

Similarly, if we suppose that β_1 is the long root of B_2 then we have

$$s_1(\beta_1) = -\beta_1 \quad s_1(\beta_2) = \beta_1 + \beta_2 \quad s_2(\beta_1) = \beta_1 + 2\beta_2 \quad s_2(\beta_2) = -\beta_2$$

and thus $(s_1 s_2)^4 = \mathrm{id}$. Then the Weyl group of B_2 is $W = D_4$.

For G_2, suppose that β_1 is the long root again. The following computation

$$s_1(\beta_1) = -\beta_1 \quad s_1(\beta_2) = \beta_1 + \beta_2 \quad s_2(\beta_1) = \beta_1 + 3\beta_2 \quad s_2(\beta_2) = -\beta_2$$

implies that $(s_1 s_2)^6 = \mathrm{id}$ and thus the Weyl group of G_2 is $W = D_6$. ■

Problem 2

Let $\Delta \subset \mathbb{R}^n$ be a root system and let $B = \{\beta_1, \dots, \beta_n\}$ be a base of Δ. The Cartan matrix is defined as

$$\left(\frac{2(\beta_i|\beta_j)}{(\beta_j|\beta_j)} \right)_{1 \le i,j \le n}.$$

1. Prove that the Cartan matrix of a root system is symmetrizable positive definite.
2. Find all Cartan matrices of rank 3.
3. Show that a Cartan matrix determines the root system uniquely up to isomorphism.

Solution: (1) We let $A = (\beta_i|\beta_j)_{1\leq i,j\leq n}$ and $x = \sum_i a_i\beta_i$ be an arbitrary nonzero vector in $\mathbb{R}^n$. Then the following result

$$x^T Ax = \left(\sum_i a_i\beta_i \mid \sum_i a_i\beta_i\right) > 0$$

implies that A is positive definite.

(2) Suppose the Dynkin diagram is disconnected. Equivalently, we say the root system is reducible. Then the there are two possible cases of Dynkin diagram:

1. Totally disconnected: $\Delta = \Delta_1 \sqcup \Delta_2 \sqcup \Delta_3$ where Δ_i is the 1-dimensional root system.
2. Partially disconnected: $\Delta = \Delta_1 \sqcup \Delta_2$ where Δ_1 is the 1-dimensional root system and Δ_2 be the irreducible 2-dimensional root system.

In the totally disconnected case, the Cartan matrix is

$$A = 2I_3 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

In the case with two connected components, after reordering the simple roots, the Cartan matrix has block diagonal form

$$A = (2) \oplus A_{\Delta_2}.$$

Since the irreducible rank-two root systems are of type A_2, $B_2 = C_2$, and G_2, the possible Cartan matrices up to simultaneous permutation of the simple roots are

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -2 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -3 & 2 \end{pmatrix}.$$

Now we suppose that Δ is an irreducible rank 3 root system. Then we have two possible cases

1. Any two roots are not orthogonal.
2. Exists two roots are orthogonal.

For case 1, we can suppose that the Cartan matrix is

$$A = \begin{pmatrix} 2 & -a_{12} & -a_{13} \\ -a_{21} & 2 & -a_{23} \\ -a_{31} & -a_{32} & 2 \end{pmatrix}$$

where $a_{ij} > 0$. The determinant of A is

$$\det(A) = 8 - 2(a_{12}a_{21} + a_{13}a_{31} + a_{23}a_{32}) - (a_{12}a_{23}a_{31} + a_{13}a_{32}a_{21}) \leq 0.$$

However, (1) implies that $\det(A) > 0$ which is a contradiction. Therefore, the case 1 is impossible. Consider the case 2. By permutation of the simple roots we can suppose the Cartan matrix be the form of

$$A = \begin{pmatrix} 2 & -a_{12} & 0 \\ -a_{21} & 2 & -a_{23} \\ 0 & -a_{32} & 2 \end{pmatrix}$$

and with determinant

$$\det(A) = 8 - 2(a_{12}a_{21} + a_{23}a_{32})$$

Since we reuqire $\det(A) > 0$, it forces

$$0 < a_{12}a_{21} + a_{23}a_{32} < 4.$$

We can suppose that $a_{12}a_{21} \leq a_{23}a_{32}$ by permutation of simple roots. For any pair of distinct simple roots, we have

$$a_{ij}a_{ji} \in \{0, 1, 2, 3\}.$$

Since we suppose the root system Δ is irreducible, equivalently, both $a_{12}a_{21}$ and $a_{23}a_{32}$ are not equal to zero. It implies that

$$(a_{12}a_{21}, a_{23}a_{32}) = (1, 1) \text{ or } (1, 2).$$

which the Cartan matrix of this two case corresponds to A_2 and $\{\mathrm{B}_2, \mathrm{C}_2\}$. Therefore, the Cartan matrix of irreducible rank 3 root system has two possible form

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -1 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -2 & 2 \end{pmatrix}.$$

(3) Let $\Delta \subset V$ and $\Delta' \subset V'$ be two root systems with bases

$$B = \{\beta_1, \ldots, \beta_n\}, \qquad B' = \{\beta_1', \ldots, \beta_n'\}.$$

Assume that they have the same Cartan matrix

$$A = (a_{ij}).$$

Define a linear map

$$T : V \longrightarrow V'$$

by

$$T(\beta_i) = \beta_i'.$$

Since the simple roots form a basis of V, this determines T uniquely.

Let s_j be the reflection corresponding to β_j, and let s_j' be the reflection corresponding to β_j'. By the definition of the Cartan matrix, we have

$$s_j(\beta_i) = \beta_i - a_{ij}\beta_j.$$

Similarly,

$$s_j'(\beta_i') = \beta_i' - a_{ij}\beta_j'.$$

Hence

$$T(s_j(\beta_i)) = T(\beta_i - a_{ij}\beta_j) = \beta_i' - a_{ij}\beta_j' = s_j'(\beta_i') = s_j'(T(\beta_i)).$$

Since this holds for every basis vector β_i, we get

$$T \circ s_j = s_j' \circ T$$

for every j.

Let W and W' be the Weyl groups of Δ and Δ'. The above identity implies that T identifies the action of W with the action of W'. Therefore, for every word $w = s_{i_1} \cdots s_{i_k} \in W$, we have

$$T(w\beta_i) = w'\beta_i',$$

where $w' = s_{i_1}' \cdots s_{i_k}' \in W'$.

Now use the standard fact that every root is obtained from a simple root by the action of the Weyl group:

$$\Delta = W \cdot B, \qquad \Delta' = W' \cdot B'.$$

It follows that

$$T(\Delta) = \Delta'.$$

Thus T is an isomorphism of root systems. Hence the Cartan matrix determines the root system uniquely up to isomorphism.

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